

AKILA ISURU KAHINGALA

Full-Stack Developer | Software Engineer

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PROFESSIONAL SUMMARY

Full Stack Developer specialized in building scalable enterprise systems and integrating advanced machine learning models into modern web applications. Recognized as the Batch Third (1st Semester) at IJSE, I have expanded my expertise through a certified AI/ML program covering LLMs, RAG pipelines, PyTorch, and deep learning. Proficient in React, Next.js, Node.js, and Java Spring Boot, I am passionate about bridging the gap between clean software architecture and deliver intelligent, secure, and seamless user experiences.

TECHNICAL SKILLS

Frontend: React 18, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap, Vite

Backend: Node.js, Express.js, Java, Spring Boot, Spring Security, Spring Data JPA, RESTful APIs

Mobile & Design: React Native (Expo), Flutter & Dart, Figma,

Databases & Cloud: MySQL, MongoDB Atlas, Cloudinary, Vercel, Firebase

AI & Machine Learning: Python, PyTorch, Google Gemini API, Pandas, NumPy,

KEY PROJECTS

LankaTravel – Luxury Travel Booking Platform — *React, TypeScript, Node.js, Express, MongoDB, Gemini API, Tailwind CSS, PayHere*

- Developed a full-featured Sri Lanka tourism platform handling user authentication, destination management, and stay bookings.
- Integrated Google Gemini API to build a personalized AI trip planner with fuzzy destination matching and structured itinerary generation.
- Implemented secure JWT authentication with refresh tokens and role-based access control (User/Admin), along with Google and Facebook OAuth.
- Configured OSRM-based travel time routing via interactive Leaflet maps and automated payment processing using the PayHere sandbox.

SkillSwap Hub – Collaborative Skill Exchange Network — *Java, Spring Boot, Spring Security (JWT), MySQL, JavaScript, HTML/CSS*

- Engineered a full-stack platform bridging users to exchange professional skills with a dual-role 'Teach/Learn' architecture.
- Designed the backend architecture utilizing Spring Boot and Spring Data JPA for efficient user, skill, and swap-request management.
- Implemented a transparent review and rating system featuring a dynamic contributor leaderboard based on community performance.
- Secured the platform with Spring Security and JWT, ensuring data integrity and role-based access for system administrators.

FairAge – Fairness-Aware Age Estimation — *Python, PyTorch, Streamlit*

- Fine-tuned a ResNet-50 model on a dataset of 86,744 images across 9 age categories, achieving over 57% test accuracy.

- Conducted a comprehensive fairness audit to identify racial accuracy gaps and applied bias-mitigation strategies including WeightedRandomSampler and class-weighted cross-entropy.
- Deployed the machine learning model via a live Streamlit web application featuring real-time prediction and probability visualization.

EDUCATION & CERTIFICATIONS

Institute of Software Engineering (IJSE) *Graduate Diploma in Software Engineering* | Sept 2024 – Present

- **Focus:** Major in Software Development | Minor in Machine Learning
- **Relevant Coursework:** Advanced Mobile App Development, Full Stack Web Development, Database Systems, Software Architecture

Institute of Software Engineering (IJSE) *Certified AI & ML Engineer Program* | Feb 2024 – Jun 2026

- **Key Topics:** Python for Machine Learning, LLM Fundamentals, RAG Pipelines, Data Analysis (Pandas, NumPy, Matplotlib), and Real-world Project Implementation

Kularathne College, Ambalangoda *G.C.E. Advanced Level (Mathematics Stream)* | 2023

- **Results:** C, C, C

EXPERIENCE

Freelance Full-Stack Developer | Self-Employed *Delivered production-grade web and desktop applications for commercial clients, ensuring robust backend architectures and responsive UI/UX.*

- **Co-engineered a full-stack Driving School Management Platform** using React, TypeScript, and Node.js/Express; implemented secure JWT authentication with Google OAuth and resolved complex SPA routing and CORS issues to ship a stable production deployment.
- **Developed an Enterprise Point-of-Sale (POS) Desktop System** leveraging a layered Java Spring Boot architecture and MySQL; designed a modern JavaFX user interface for real-time stock tracking, order management, and daily sales reporting.
- **Managed end-to-end project lifecycles**, translating initial client requirements into scalable database models (MongoDB Atlas/MySQL) and responsive frontends, while ensuring high system reliability.

REFERENCE

- **Mr. Pubudu Chamod**

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